MOBILE APPLICATION DEVELOPMENT

SYLLABUS

UNIT I: **Introduction to Android**

UNIT II: **Android Application Design Essentials**

UNIT III: **Android User Interface Design Essentials**

UNIT IV: **Using Common Android APIs**

Unit I: Introduction to Android

1. The Android Platform

- Overview of Android OS
- Features of Android
- Android architecture (Linux Kernel, Libraries, Android Runtime, Application Framework)
- Android versions and API levels

2. Android SDK (Software Development Kit)

- Overview of Android SDK components
- Android Studio vs. Eclipse (historical usage)
- Key tools: ADB (Android Debug Bridge), Emulator, SDK Manager

3. Eclipse Installation (Legacy)

- Eclipse IDE setup (Android Development Tools - ADT)
- Configuring Android SDK in Eclipse
- Running an Android project in Eclipse

4. Android Studio Installation (Current)

- Installing Android Studio
- Setting up SDK, Virtual Device (AVD)
- Understanding project structure

5. Building Your First Android Application

- Creating a new project in Android Studio
- Understanding project folders (`res`, `manifest`, `java`, `gradle`)
- Running the app on an emulator or real device

6. Understanding Anatomy of an Android Application

- Components of an Android app:
  - **Activities** (UI screens)
  - **Services** (background processes)
  - **Broadcast Receivers** (listening to system events)
  - **Content Providers** (data sharing)
- The role of `MainActivity.java`
- `R.java` and resource handling

7. Android Manifest File

- What is `AndroidManifest.xml`?
- Declaring app permissions
- Defining application components (Activities, Services, Receivers)
- Setting the launcher activity

What is the Manifest File in Android?

The **AndroidManifest.xml** file is a critical configuration file in every Android application. It provides essential information about the app to the **Android system** so it can run correctly.

Key Roles of AndroidManifest.xml

✅ Declares the **package name** (unique identifier for the app)
✅ Registers **app components** (Activities, Services, Broadcast Receivers, Content Providers)
✅ Defines **permissions** (e.g., Internet, Camera, Location)
✅ Specifies **minimum & target SDK versions**
✅ Configures **hardware and software features**

Gradle is the build system used in Android development to automate the process of compiling, linking, and packaging Android applications. It manages dependencies, builds APK/AAB files, and optimizes project configurations.

Unit II: Android Application Design Essentials

1. Anatomy of an Android Application

- Structure of an Android project
- `res/` folder (layouts, drawables, values)
- `AndroidManifest.xml` and Gradle files
- `src/` folder (Java/Kotlin code)

2. Android Terminologies

- **Activity**: A single screen with UI
- **Intent**: Used to navigate between screens or start services
- **Service**: Background tasks without UI
- **Content Provider**: Shares data between apps
- **Broadcast Receiver**: Listens for system or custom events
- **Fragment**: Reusable UI component within an activity
- **ViewModel**: Holds UI-related data across activity lifecycle

3. Application Context

- Difference between **Application Context** and **Activity Context**
- When to use each type (e.g., Singleton patterns, database access)

4. Activities

- Lifecycle of an activity (`onCreate()`, `onResume()`, `onPause()`, etc.)
- Navigating between activities using **Intents**
- Passing data between activities

5. Services

- Types of Services:
  - **Foreground Service** (e.g., music player)
  - **Background Service** (e.g., fetching data)
  - **Bound Service** (e.g., communication with activity)
- Lifecycle methods (`onStartCommand()`, `onDestroy()`)

6. Intents

- **Explicit Intents**: Used for communication between components of the same app
- **Implicit Intents**: Used to communicate with system apps or other apps

7. Receiving and Broadcasting Intents

- **Broadcast Receivers**: Listening for system broadcasts (e.g., battery low)
- **Custom Broadcasts**: Sending your own broadcasts to other components

8. Android Manifest File and Its Common Settings

- Declaring activities, services, receivers
- Setting permissions
- Specifying app theme and launcher activity

9. Using Intent Filters

- What are Intent Filters?
- Adding an Intent Filter in `AndroidManifest.xml`
- Handling custom actions

10. Permissions in Android

- Types of permissions:
  - **Normal Permissions** (Internet, Bluetooth)
  - **Dangerous Permissions** (Camera, Location)
- Requesting runtime permissions in Android 6.0+ (`requestPermissions()`)

Unit III: Android User Interface Design Essentials

1. **User Interface Screen Elements**
   - Views and ViewGroups
   - Buttons, TextView, EditText, ImageView, CheckBox, RadioButton, ProgressBar, etc.
   - RecyclerView and ListView for displaying lists
2. **Designing User Interfaces with Layouts**
   - LinearLayout, RelativeLayout, ConstraintLayout, FrameLayout, TableLayout
   - Nested Layouts and best practices for UI efficiency
3. **Drawing and Working with Animation**
   - Custom drawing with Canvas and Paint
   - Frame-by-frame animations
   - Property animations (ObjectAnimator, ValueAnimator)
   - MotionLayout for advanced animations
4. **Testing Android Applications**
   - Unit Testing (JUnit)
   - UI Testing (Espresso)
   - Debugging tools in Android Studio
   - Performance profiling

5. Publishing Android Applications

- Creating a signed APK/AAB
- Play Store guidelines and submission process
- Versioning and updates

6. Using Android Preferences

- SharedPreferences for storing small amounts of data
- PreferenceScreen and PreferenceFragment for user settings

7. Managing Application Resources in a Hierarchy

- res folder structure: drawable, layout, values (strings, colors, dimens)
- Localization for different languages (strings.xml)
- Supporting multiple screen sizes and densities (drawable-hdpi, xhdpi, etc.)

8. Working with Different Types of Resources

- XML drawables (shape, selector, layer-list)
- Vector drawables and animations
- String, color, and dimension resources

Unit IV: Using Common Android APIs

1. **Using Android Data and Storage APIs**
   - Internal storage (files directory)
   - External storage (SD card, MediaStore)
   - Data backup and cloud storage
2. **Managing Data using SQLite**
   - SQLite database creation and management
   - Using Room Database (modern approach)
   - CRUD operations in SQLite
3. **Sharing Data Between Applications with Content Providers**
   - Creating a Content Provider
   - Accessing data from another app using a Content Resolver
   - Permissions for secure data sharing
4. **Using Android Networking APIs**
   - HTTP requests with Retrofit & Volley
   - Parsing JSON responses
   - WebSockets and real-time communication

5. Using Android Web APIs

- WebView for embedding web content
- Calling RESTful APIs
- OAuth authentication for third-party APIs

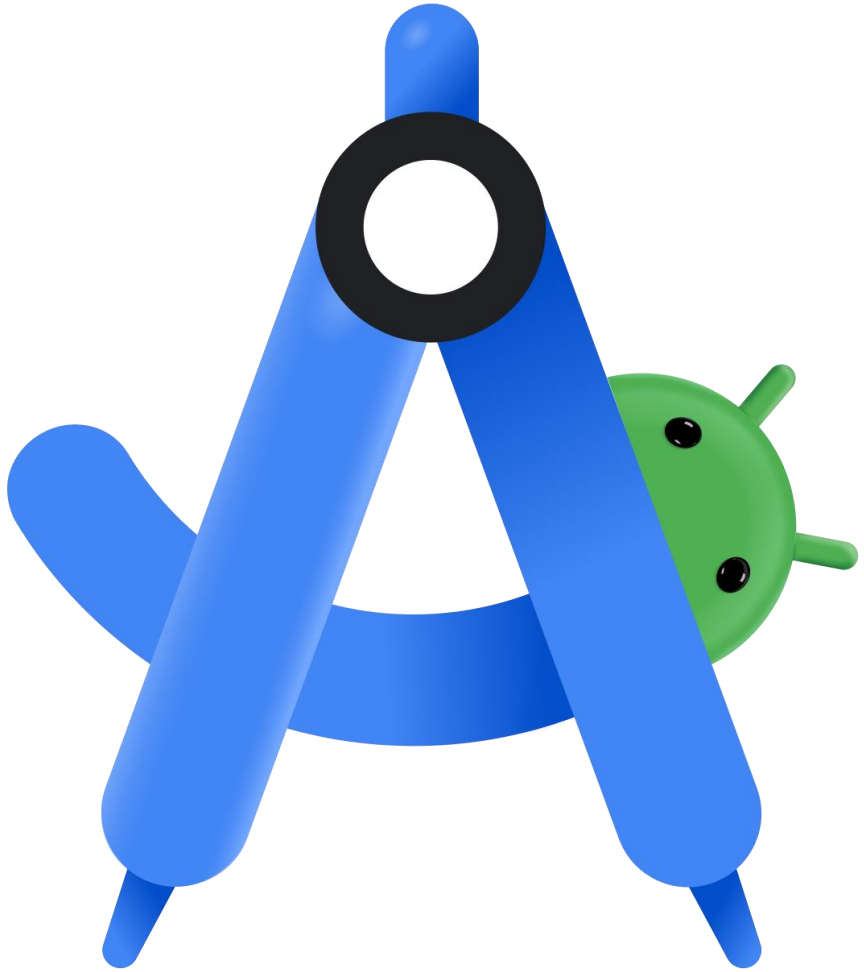
6. Using Android Telephony APIs

- Sending SMS programmatically
- Detecting incoming calls
- Accessing SIM card details

7. Deploying Android Applications to the World

- App distribution on Google Play Store
- Alternative distribution methods (APK sharing, third-party stores)
- Firebase App Distribution for beta testing

Android Studio



Android Studio

- Android Studio is the official **Integrated Development Environment (IDE)** for Android app development, developed by **Google** and based on **JetBrains' IntelliJ IDEA**. It provides a complete development environment for building, testing, and debugging Android applications.

Android Studio

- Android Studio was officially announced (transition from Eclipse) as the **official Integrated Development Environment (IDE) for Android app development** by **Google** on **May 16, 2013**, at the **Google I/O 2013** conference.
- **In 2015** Support for Eclipse ADT (Android Development Tools) officially ended.

Download and Installation of Android Studio

https://developer.android.com/studio/archive

Download and Installation of Android Studio

Basic System Requirements:

Android Studio system requirements

The following are the system requirements for Android Studio on Windows.

- 64-bit Microsoft® Windows® 8/10/11
- x86_64 CPU architecture; 2nd generation Intel Core or newer, or AMD CPU with support for a Windows Hypervisor
- 8 GB RAM or more
- 8 GB of available disk space minimum (IDE + Android SDK + Android Emulator)
- 1280 x 800 minimum screen resolution

Download and Installation of Android Studio

Older versions of Android Studio requires less resources.

Hence, you can download the previous versions from given link below.

Older versions: https://developer.android.com/studio/archive
Latest version: https://developer.android.com/studio

Download Older Versions of Android Studio

Android Studio download archi

developer.android.com/studio/archive

Developers | Essentials | Design & Plan | Develop | Google Play | Community | Search / | English | Android Studio | Sign in

ANDROID STUDIO

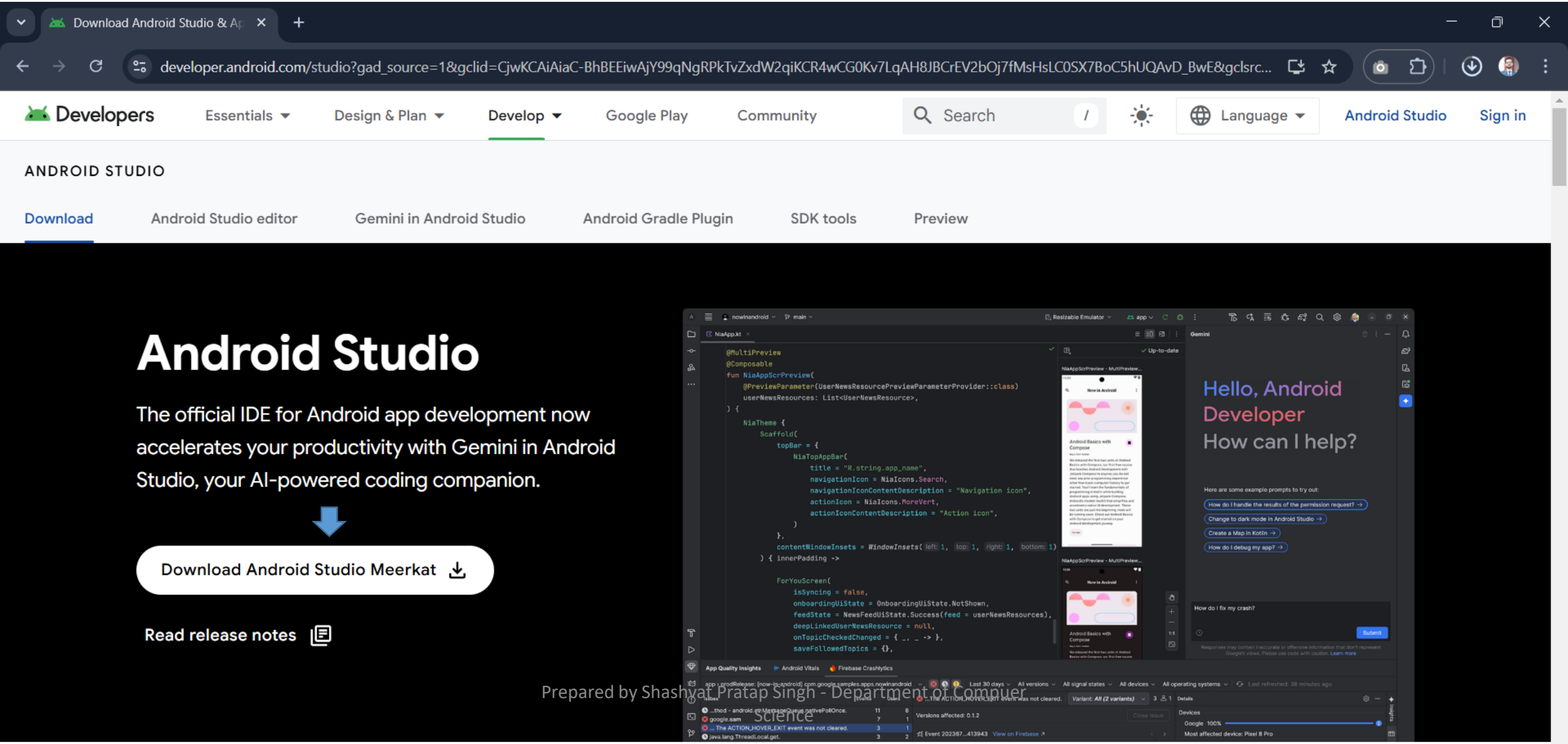
Android Studio download archives

This page provides an archive of Android Studio releases. However, we recommend that you download the latest stable version or the latest preview version.

For Android Emulator downloads, see the Emulator download archives.

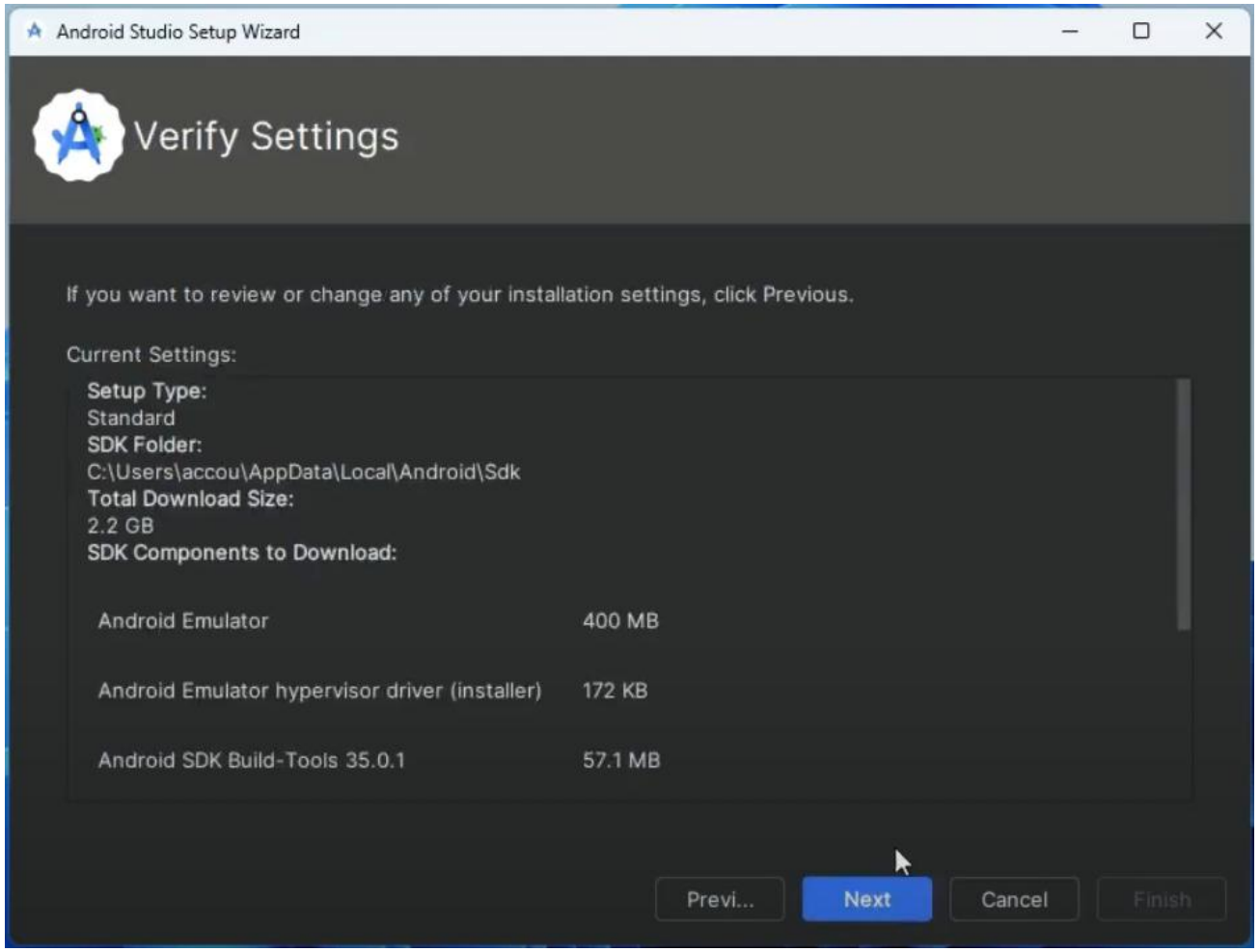
- Android Studio Meerkat | 2024.3.1 March 4, 2025
- Android Studio Meerkat Feature Drop | 2024.3.2 Canary 7 February 27, 2025
- Android Studio Ladybug Feature Drop | 2024.2.2 Patch 2 February 26, 2025
- Android Studio Meerkat Feature Drop | 2024.3.2 Canary 6 February 21, 2025
- Android Studio Meerkat | 2024.3.1 RC 2 February 18, 2025
- Android Studio Meerkat Feature Drop | 2024.3.2 Canary 5 February 13, 2025
- Android Studio Ladybug Feature Drop | 2024.2.2 Patch 1 February 13, 2025

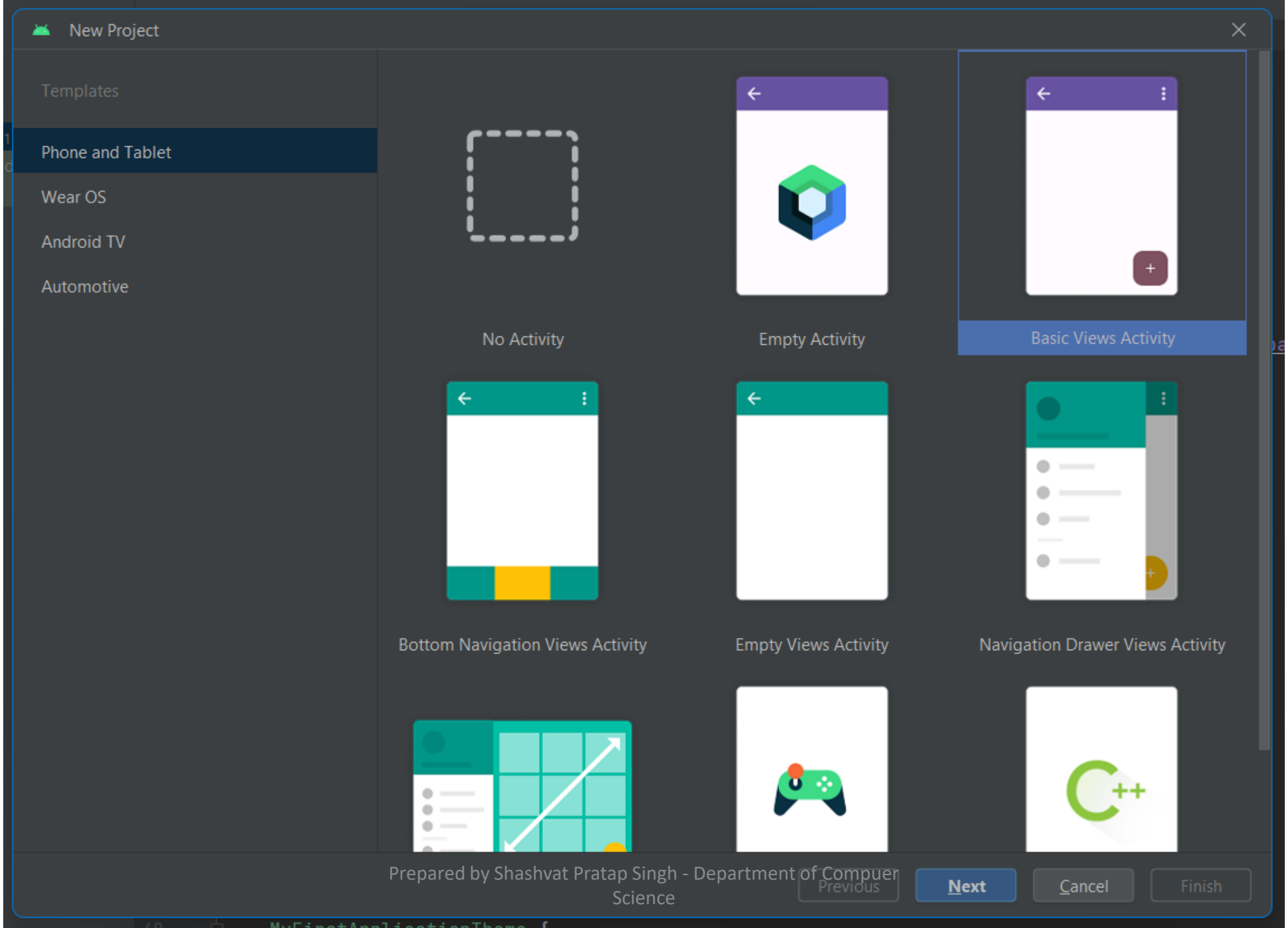
Download Latest Version of Android Studio (1.2GB)



Download Latest Version of Android Studio (1.2GB)

Note: After downloading the Android Studio (exe), it will take around 2GB to download necessary SDK components during setup.




New Project
Templates
Phone and Tablet
Wear OS
Android TV
Automotive
No Activity
Empty Activity
Basic Views Activity
Bottom Navigation Views Activity
Empty Views Activity
Navigation Drawer Views Activity
Previous
Next
Cancel
Finish

New Project

Basic Views Activity

Creates a new basic activity

Name: My Application

Package name: com.example.myapplication

Save location: C:\Users\shash\StudioProjects\MyApplication2

Language: Kotlin

Minimum SDK: API 26: Android 8.0 (Oreo)

Your app will run on approximately **95.4%** of devices.

Help me choose

Previous | Next | Cancel | Finish

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help My First Application - C:\Users\shash\StudioProjects\MyFirstApplication\app\src\main\java\com\example\myfirstapplication\MainActivity.kt - Administrator

MyFirstApplication 〉 app 〉 src 〉 main 〉 java 〉 com 〉 example 〉 myfirstapplication 〉 MainActivity.kt

Add Configuration... Pixel 4a API 32

Android

loading...

MainActivity.kt

Gradle project sync in progress...

Indexing...

```kotlin
package com.example.myfirstapplication

import ...

class MainActivity : ComponentActivity() {
    override fun onCreate(savedInstanceState: Bundle?) {
        super.onCreate(savedInstanceState)
        setContent {
            MyFirstApplicationTheme {
                // A surface container using the 'background' color from the theme
                Surface(modifier = Modifier.fillMaxSize(), color = MaterialTheme.colorScheme.background) {
                    Greeting( name: "Android")
                }
            }
        }
    }
}

@Composable
fun Greeting(name: String, modifier: Modifier = Modifier) {
    Text(
        text = "Hello $name!",
        modifier = modifier
    )
}

@Preview(showBackground = true)
@Composable
fun GreetingPreview() {
    MyFirstApplicationTheme {
        Greeting( name: "Android")
    }
```

Do not disturb the IDE until **“Gradle Build”** is in process

Prepared by Shashvat Pratap Singh - Department of Compuer Science

Version Control TODO Problems Terminal App Inspection Logcat App Quality Insights Services Build

Gradle sync started (4 minutes ago) Gradle: Build model... Show all (2) 1:1 LF UTF-8 4 spaces

File Structure of Android Project

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help Hello Android - MainActivity.java [Hello_Android.app.main] - Administrator

HelloAndroid 〉 app 〉 src 〉 main 〉 res 〉 layout 〉 activity_main.xml app Pixel 4a API 32

Android

- app
  - manifests
    - AndroidManifest.xml 3/6/2025 12:55 AM, 1.06 kB
  - java
    - com.example.helloandroid
      - FirstFragment 3/6/2025 12:55 AM, 1.28 kB
      - MainActivity 3/6/2025 12:55 AM, 2.66 kB 2 minutes ago
      - SecondFragment 3/6/2025 12:55 AM, 1.29 kB
    - com.example.helloandroid (androidTest)
      - ExampleInstrumentedTest 3/6/2025 12:55 AM, 762 B
    - com.example.helloandroid (test)
      - ExampleUnitTest 3/6/2025 12:55 AM, 385 B
  - res
    - drawable
    - layout
      - activity_main.xml 3/6/2025 12:55 AM, 1.36 kB A minute ag
      - content_main.xml 3/6/2025 12:55 AM, 907 B
      - fragment_first.xml 3/6/2025 12:55 AM, 1.56 kB A minute ag
      - fragment_second.xml 3/6/2025 12:55 AM, 1.57 kB Momen
    - menu
    - mipmap
    - navigation
    - values
    - xml
- Gradle Scripts

MainActivity.java | activity_main.xml | fragment_first.xml | fragment_second.xml

```
package com.example.helloandroid;

import android.os.Bundle;

import com.google.android.material.snackbar.Snackbar;

import androidx.appcompat.app.AppCompatActivity;

import android.view.View;

import androidx.core.view.WindowCompat;
import androidx.navigation.NavController;
import androidx.navigation.Navigation;
import androidx.navigation.ui.AppBarConfiguration;
import androidx.navigation.ui.NavigationUI;

import com.example.helloandroid.databinding.ActivityMainBinding;

import android.view.Menu;
import android.view.MenuItem;

3 usages
public class MainActivity extends AppCompatActivity {
```

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help Hello Android - activity_main.xml [Hello_Android.app.main] - Administrator

HelloAndroid › app › src › main › res › layout › activity_main.xml

app | Pixel 4a API 32

MainActivity.java | content_main.xml | activity_main.xml | fragment_first.xml | fragment_second.xml

Code | Split | Design

Android

- app
  - manifests
    - AndroidManifest.xml 3/6/2025 12:55 AM, 1.06 kB
  - java
    - com.example.helloandroid
      - FirstFragment 3/6/2025 12:55 AM, 1.28 kB
      - MainActivity 3/6/2025 12:55 AM, 2.66 kB A minute ago
      - SecondFragment 3/6/2025 12:55 AM, 1.29 kB
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    - com.example.helloandroid (test)
      - ExampleUnitTest 3/6/2025 12:55 AM, 385 B
  - res
    - drawable
    - layout
      - activity_main.xml 3/6/2025 12:55 AM, 1.36 kB Moments ag
      - content_main.xml 3/6/2025 12:55 AM, 907 B Moments ago
      - fragment_first.xml 3/6/2025 12:55 AM, 1.56 kB Moments a
      - fragment_second.xml 3/6/2025 12:55 AM, 1.57 kB 3 minut
    - menu
    - mipmap
    - navigation
    - values
    - xml
  - Gradle Scripts

Palette

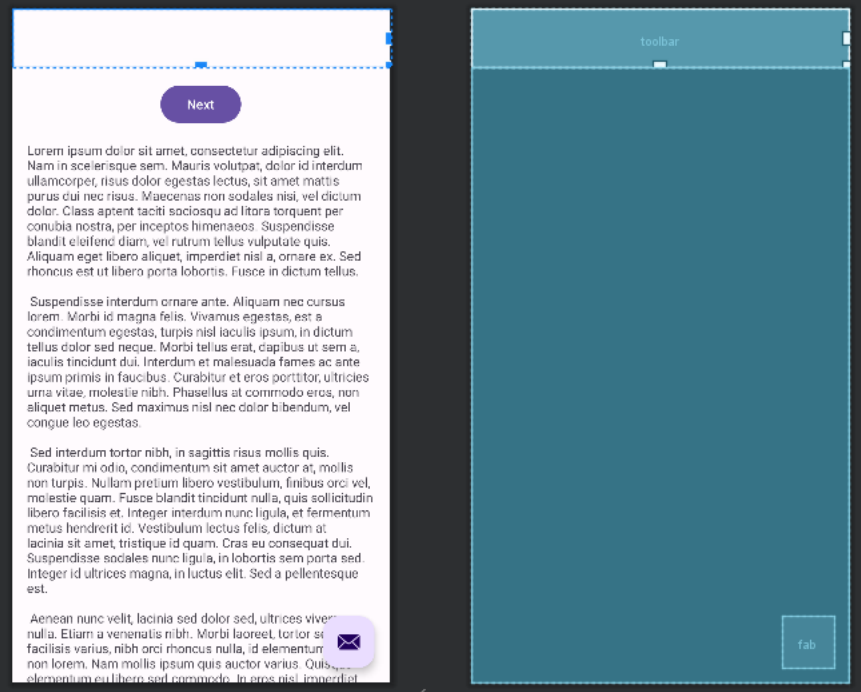
- Common
- Text
- Buttons
- Widgets
- Layouts
- Containers
- Helpers
- Google
- Legacy

Ab TextView, Button, ImageView, RecyclerView, FragmentCon..., ScrollView, Switch

Component Tree

- CoordinatorLayout
  - AppBarLayout
    - toolbar
  - <include> @layout/content...
  - fab

activity_main.xml | Pixel | 34 | HelloAndroid | Default (en-us)



Attributes

MaterialToolbar toolbar

id: toolbar

Declared Attributes

Layout

- layout_width: match_parent
- layout_height: ?attr/actionBarSize
- visibility
- visibility

Transforms

View

Rotation

- x 0
- y 0
- z 0

- rotation
- rotationX
- rotationY
- scaleX
- scaleY
- translationX

androidx.coordinatorlayout.widget.CoordinatorLayout › com.google.android.material.appbar.AppBarLayout › com.google.android.material.appbar.MaterialToolbar

Prepared by Shashwat Pratap Singh Department of Computer Science

Version Control | TODO | Problems | Terminal | App Inspection | Logcat | App Quality Insights | Services | Build | Profiler | Layout Inspector

Gradle sync finished in 15 s 962 ms (5 minutes ago)

LF UTF-8 4 spaces

Run your Android Application

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help Hello Android - MainActivity.java [Hello_Android.app.main] - Administrator

HelloAndroid 〉 app 〉 src 〉 main 〉 res 〉 layout 〉 activity_main.xml

app | Pixel 4a API 32

Run

Device Manager

Android

- app
  - manifests
    - AndroidManifest.xml 3/6/2025 12:55 AM, 1.06 kB
  - java
    - com.example.helloandroid
      - FirstFragment 3/6/2025 12:55 AM, 1.28 kB
      - MainActivity 3/6/2025 12:55 AM, 2.66 kB 2 minutes ago
      - SecondFragment 3/6/2025 12:55 AM, 1.29 kB
    - com.example.helloandroid (androidTest)
      - ExampleInstrumentedTest 3/6/2025 12:55 AM, 762 B
    - com.example.helloandroid (test)
      - ExampleUnitTest 3/6/2025 12:55 AM, 385 B
  - res
    - drawable
    - layout
      - activity_main.xml 3/6/2025 12:55 AM, 1.36 kB A minute ag
      - content_main.xml 3/6/2025 12:55 AM, 907 B
      - fragment_first.xml 3/6/2025 12:55 AM, 1.56 kB A minute ag
      - fragment_second.xml 3/6/2025 12:55 AM, 1.57 kB Momen
    - menu
    - mipmap
    - navigation
    - values
    - xml
- Gradle Scripts

MainActivity.java | activity_main.xml | fragment_first.xml | fragment_second.xml

```
1  package com.example.helloandroid;
2
3  import android.os.Bundle;
4
5  import com.google.android.material.snackbar.Snackbar;
6
7  import androidx.appcompat.app.AppCompatActivity;
8
9  import android.view.View;
10
11 import androidx.core.view.WindowCompat;
12 import androidx.navigation.NavController;
13 import androidx.navigation.Navigation;
14 import androidx.navigation.ui.AppBarConfiguration;
15 import androidx.navigation.ui.NavigationUI;
16
17 import com.example.helloandroid.databinding.ActivityMainBinding;
18
19 import android.view.Menu;
20 import android.view.MenuItem;
21
   3 usages
22 public class MainActivity extends AppCompatActivity {
```

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help Hello Android - activity_main.xml [Hello_Android.app.main] - Administrator

HelloAndroid › app › src › main › res › layout › activity_main.xml

app | Pixel 4a API 32

Android

- app
  - manifests
    - AndroidManifest.xml 3/6/2025 1:04 AM, 1.06 kB
  - java
    - com.example.helloandroid
      - FirstFragment 3/6/2025 1:04 AM, 1.27 kB
      - MainActivity 3/6/2025 1:04 AM, 2.64 kB 9 minutes ago
      - SecondFragment 3/6/2025 1:04 AM, 1.27 kB
    - com.example.helloandroid (androidTest)
      - ExampleInstrumentedTest 3/6/2025 12:55 AM, 762 B
    - com.example.helloandroid (test)
      - ExampleUnitTest 3/6/2025 12:55 AM, 385 B
  - java (generated)
  - res
    - drawable
    - layout
      - activity_main.xml 3/6/2025 1:04 AM, 1.33 kB 6 minutes ago
      - content_main.xml 3/6/2025 1:04 AM, 911 B 8 minutes ago
      - fragment_first.xml 3/6/2025 1:04 AM, 1.49 kB 8 minutes ag
      - fragment_second.xml 3/6/2025 1:04 AM, 1.5 kB 11 minute
    - menu
    - mipmap
    - navigation
    - values
    - xml
  - res (generated)
- Gradle Scripts

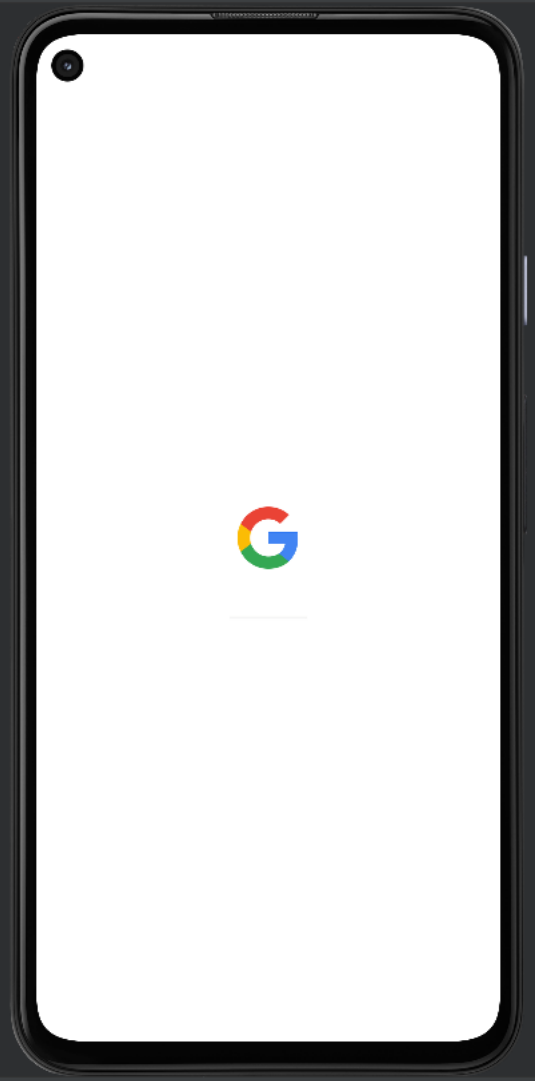
MainActivity.java | content_main.xml | activity_main.xml | fragment_first.xml

Code | Split | Design

```
10
11  <com.google.android.material.appbar.AppBarLayout
12      android:layout_height="wrap_content"
13      android:layout_width="match_parent"
14      android:fitsSystemWindows="true"
15      >
16
17      <com.google.android.material.appbar.MaterialToolbar
18          android:id="@+id/toolbar"
19          android:layout_width = "match_parent"
20          android:layout_height = "?attr/actionBarSize" />
21
22  </com.google.android.material.appbar.AppBarLayout>
23
24  <include layout="@layout/content_main"/>
25
26  <com.google.android.material.floatingactionbutton.FloatingActionB
27  android:id="@+id/fab"
28  android:layout_width="wrap_content"
29  android:layout_height="wrap_content"
30  android:layout_gravity="bottom|end"
31  android:layout_marginEnd="16dp"
32  android:layout_marginBottom="16dp"
33  app:srcCompat="@android:drawable/ic_dialog_email" />
34
35  </androidx.coordinatorlayout.widget.CoordinatorLayout>
```

androidx.coordinatorlayout.widget.CoordinatorLayout › com.google.android.material.appbar.AppBarLayout

Running Devices: Pixel 4a API 32



Run: app

```
03/06 01:09:11: Launching 'app' on Pixel 4a API 32.
```

Prepared by Shashvat Pratap Singh - Department of Compuer Science

Version Control | Run | TODO | Problems | Terminal | App Inspection | Logcat | App Quality Insights | Services | Build | Profiler | Layout Inspector

Gradle build finished in 1 s 194 ms (moments ago) Waiting for all target devices to come online 17:22 4 spaces

File Edit View Navigate Code Refactor Build Run Tools VCS Window Help Hello Android - activity_main.xml [Hello_Android.app.main] - Administrator

HelloAndroid 〉 app 〉 src 〉 main 〉 res 〉 layout 〉 activity_main.xml

app | Pixel 4a API 32

Android

- app
  - manifests
    - AndroidManifest.xml 3/6/2025 1:04 AM, 1.06 kB
  - java
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  - java (generated)
  - res
    - drawable
    - layout
      - activity_main.xml 3/6/2025 1:04 AM, 1.33 kB 6 minutes ago
      - content_main.xml 3/6/2025 1:04 AM, 911 B 9 minutes ago
      - fragment_first.xml 3/6/2025 1:04 AM, 1.49 kB 9 minutes ag
      - fragment_second.xml 3/6/2025 1:04 AM, 1.5 kB 12 minute
    - menu
    - mipmap
    - navigation
    - values
    - xml
  - res (generated)
- Gradle Scripts

MainActivity.java | content_main.xml | activity_main.xml | fragment_first.xml

Code | Split | Design

Palette

- Common
- Text
- Buttons
- Widgets
- Layouts
- Containers
- Helpers
- Google
- Legacy

Ab TextView
Button
ImageView
RecyclerView
FragmentCon...
ScrollView
Switch

Component Tree

- CoordinatorLayout
  - AppBarLayout
    - toolbar
  - <include> @layout/content...
  - fab

activity_main.xml | Pixel

Next

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androidx.coordinatorlayout.widget.CoordinatorLayout 〉 com.google.android.material.appbar.AppBarLayout

Running Devices: Pixel 4a API 32

1:10

First Fragment

Next

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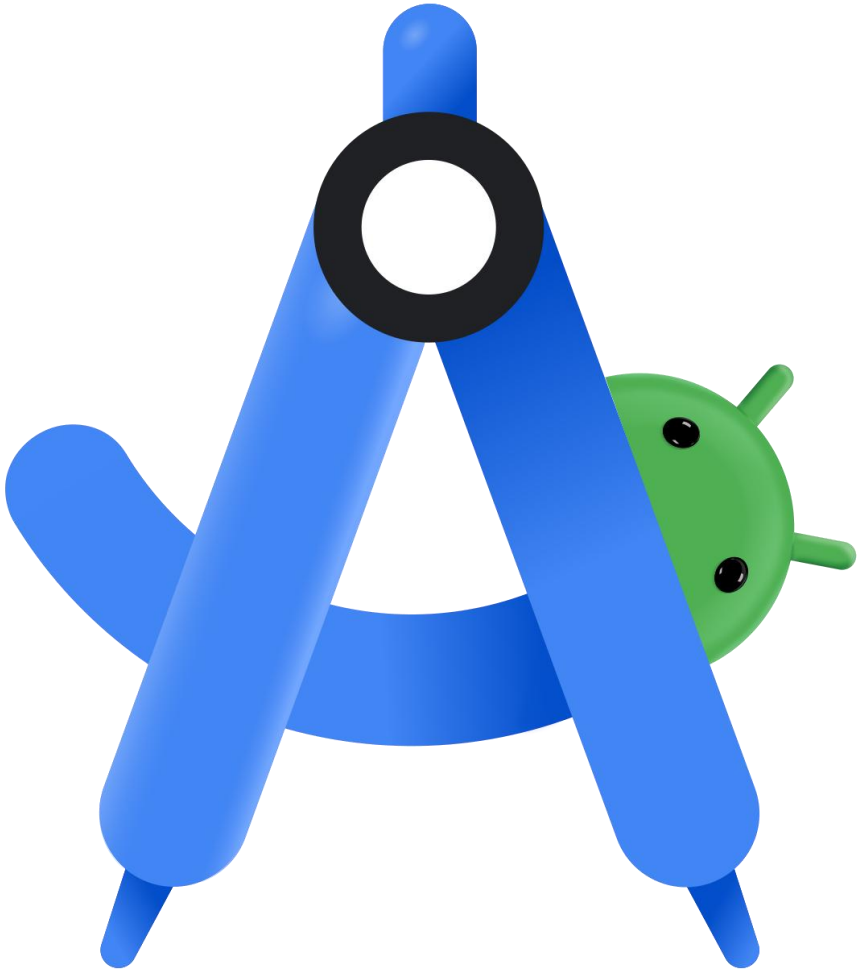
Run: app

Connected to process 2305 on device 'Pixel_4a_API_32 [emulator-5554]'.

Version Control | Run | TODO | Problems | Terminal | App Inspection | Logcat | App Quality Insights | Services | Build | Profiler | Layout Inspector

Launch succeeded (moments ago) 4 spaces

Prepared by Shashvat Pratap Singh - Department of Compuer Science

Thank You